Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Integration-focused Software Engineer with 2+ years building **REST APIs**, **ETL pipelines**, scheduled automation, data transformations, and production monitoring using Java, JavaScript, Groovy, SQL, AWS, and Azure.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Operating Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Built **ETL-style integrations** across FastAPI, PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental loads, reconciliation, error handling, and CI/CD-backed releases at 99%+ reliability.
- Restructured fault-tolerant asynchronous orchestration with explicit job states and failure isolation, processing 1K+ tasks while tripling throughput under concurrent workloads and preventing cascading failures across dependent services.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Designed concurrent batch processing with multithreading and secure ingestion, processing 1K+ files per batch through **REST APIs** and AWS RDS with recoverable execution and automated validation workflows.
- Implemented production services handling 10K+ daily requests for 150+ instructors, using CI/CD-driven workflows, usage tracking, and fallback execution to reduce feedback turnaround 60% during peak production load.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Designed API and **data-transformation flows** using OOP, AWS Glue, S3, JSON, CSV, and Parquet, improving response times 30% and ETL efficiency 68% while reducing technical debt 20%.
- Automated nightly VM allocation, builds, installation, integration tests, unit tests, and CSV reports through scheduled Jenkins jobs, validating end-to-end synchronization across Windows, Linux, and macOS release environments.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed production workflows.
- Built automated Docker deployments through Jenkins and Groovy CI/CD pipelines, achieving 95%+ successful validation and cutting release time 50% through staged integration testing, monitoring, and repeatable configurations.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Developed Flask APIs, payment webhooks, pricing, and coupon workflows while synchronizing Firebase booking events and processing hourly Databricks micro-batches across 50K+ records with reliable sub-200-millisecond API responses.
- Designed end-to-end workflows, mapped 12+ backend integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation and alignment.

Skills

Languages & Formats: Java, JavaScript, TypeScript, Python, Groovy, SQL, Bash, JSON, XML, YAML
Integration & APIs: REST APIs, Webhooks, JWT, WebSockets, FastAPI, Flask, Node.js, Express.js, Spring Boot, Microservices
Data & Automation: ETL Pipelines, Reconciliation Checks, Data Transformation, Batch Processing, Job Scheduling, AWS Glue, Databricks, PostgreSQL, SQL Server, Firebase
Delivery & Operations: AWS, Azure, Docker, Kubernetes, Jenkins, CI/CD, Git, Postman, Agile/Scrum, Unit Testing, Integration Testing, System Validation, Monitoring, Error Handling, Root-Cause Analysis

Projects

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a schema-aware transformation pipeline generating optimized SQL with 82% accuracy, implementing validation and structured execution across ten database engines while supporting cross-platform development through Docker and Maven.

Linguistix | *Python, AWS ECS/Fargate, Docker, CI/CD*

- Developed and deployed a containerized RESTful data-processing API on AWS ECS/Fargate with CodePipeline releases, automated validation checks, 87% measured accuracy, and reliable repeatable production deployment workflows across environments.

FinNexus Live | *TypeScript, Kubernetes, MongoDB, WebSockets*

- Built a TypeScript application supporting 50K+ users and 1M+ transactions with secure integrations, real-time synchronization, advanced filtering, and reliable containerized deployment workflows operating successfully at production scale.